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Alphabets for Quantum Field Theory

A State of the Art and Objectives

We have entered an era of precision physics. Data streams in from all distance scales, from the smallest to the largest. The Large Hadron collider will gather over an order of magnitude more data in the coming decades, while gravitational wave telescopes will go from scattered observations to the ability to statistically analyze properties of compact astrophysical objects in unprecedented detail. It is our responsibility as scientists to use the coming wealth of data as best we can, to have as accurate a picture as possible of the natural world.

On both the largest and smallest scales, an important theoretical prerequisite for precision physics is the ability to compute to high orders in perturbative quantum field theory. In recent years great strides have been made in these computations via an understanding of a class of functions called **multiple polylogarithms**. Multiple polylogarithms are iterated integrals of rational functions, and they can be used to represent a wide variety of perturbative scattering amplitudes. Every one-loop Feynman diagram can be represented in terms of multiple polylogarithms [1, 2], as can many diagrams involving massless particles. The last decade of progress has led to great strides in understanding and manipulating amplitudes expressible in multiple polylogarithms, leading to efficient identification of identities, fast-converging series expansions, well-controlled analytic continuation, and powerful differential equation methods.

These new methods have been fantastically productive, but they cannot support the field forever. Increasingly we must consider Feynman diagrams that *cannot* be expressed in terms of multiple polylogarithms, Feynman diagrams **beyond polylogarithms**. These diagrams cannot be represented in terms of iterated integrals over rational functions, but instead must be represented in terms of integrals of algebraic functions. The simplest such algebraic functions are square roots, and their arguments are polynomials defining curves or higher-dimensional spaces. Integrals of algebraic functions of this kind have growing relevance to both collider and gravitational wave physics.

In collider physics, an important upcoming goal is the precision measurement of quantities in the electroweak sector, such as the quadratic gauge coupling and the Higgs-top Yukawa coupling. Many of these quantities will be measured to the percent level by upcoming experiments, which on the theory side will require two- and in some cases three-loop calculations [3]. As the masses of electroweak gauge bosons, the Higgs, and the top quark cannot be neglected in these calculations, they will often involve diagrams that go beyond the space of multiple polylogarithms [4]. For example, a two-loop diagram that will contribute to integral bases for both quadratic gauge coupling and Higgs-top Yukawa measurements already involves an integral over a higher-dimensional space called a K3 [5, 6, 7, 8].

In gravitational wave physics, the situation is in some senses simpler. Quantum field theory-based approaches typically treat black holes and other compact objects as massive scalars, interacting through the massless graviton field [9]. The classical limit then further simplifies results. However, these approaches have even higher potential and need for precision, with interesting effects present at five loops [10]. As such, diagrams beyond polylogarithms are already relevant in state-of-the-art calculations [11], with higher-dimensional spaces are just around the corner.

The simplest spaces needed to describe functions beyond multiple polylogarithms are elliptic curves, and iterated integrals involving these are often called **elliptic multiple polylogarithms**. There is increasing progress in manipulating these functions, both from the perspective of them as iterated integrals over algebraic functions and a picture as iterated integrals on a torus [12, 13, 14, 15]. However, this progress still has not matched the capabilities we have accumulated with multiple polylogarithms, and there are still important mysteries.

If one starts with diagrams that must be expressed in elliptic multiple polylogarithms, it is often only a small step to find a diagram that cannot be expressed in even these functions, and must be expressed using a higher-dimensional space. Adding loops, additional particles, or new mass parameters all can increase the complexity of the geometry one must consider. Surprisingly often, these higher-dimensional spaces are **Calabi-Yau**. The K3 mentioned above is one of the simplest classes of Calabi-Yau, the simplest being elliptic curves themselves. Calabi-Yaus have been studied by string theorists in other contexts, as candidates to compactify extra dimensions and preserve supersymmetry. There appears to be no direct connection between this history and their appearance in Feynman diagrams, where they can easily appear even in non-supersymmetric theories. While the string theory literature contains many results regarding Calabi-Yaus, these do not directly result in useful tools for Feynman diagrams, and we must make clever use of the literature to make progress. In particular, there is not as of yet a general notion of a “Calabi-Yau polylogarithm”.

In order to efficiently manipulate and evaluate Feynman diagrams beyond polylogarithms, we will need tools comparable to those developed for the polylogarithmic case. For multiple polylogarithms, the key element that jump-started progress was the notion of a **symbol alphabet** [16]. A symbol alphabet is a formalism that keeps track of branch cuts of multiple polylogarithms in a systematic way, as "letters" in "words" that describe iterated integrals. A linear combination of these "words" describing a given function is then referred to as a **symbol**, with the full set of branch cut letters referred to as the symbol's **alphabet**. The symbol alphabet formalism allows one to find identities between functions, re-expressing one set of iterated integrals in another form such that one only needs to fix a few "beyond-the-symbol" constant terms. As an organizing principle, it inspired the method of canonical differential equations, the foundation of the current state of the art in computing Feynman integrals. By expressing a differential equation in terms of symbol letters, multiple polylogarithm solutions can be systematically constructed order by order, easily finding expressions to any required order in the ϵ expansion [17].

These tools have been immensely useful in collider physics and gravity, but they had their origin in a more abstract context: the planar limit of $\mathcal{N} = 4$ super Yang-Mills theory (or MSYM). $\mathcal{N} = 4$ super Yang-Mills is a conformal theory, and its planar limit has an additional dual conformal symmetry when amplitudes are interpreted as Wilson loops in a dual position space. These symmetries place the theory in a "sweet spot": with amplitudes simple enough to study in great generality, but complex enough to reveal useful lessons for working with other theories. The notion of a symbol alphabet was originally introduced to the physics community via research in this theory in ref. [16]. Later research found that the symbol alphabets of polylogarithmic amplitudes in MSYM have a variety of surprising properties. As amplitudes in MSYM can also go beyond polylogarithms (and in fact can involve arbitrarily high-dimension Calabi-Yaus [18]), a crucial question is whether these surprising properties persist beyond polylogarithms, and what form they take. This question is important not merely for researchers in supersymmetric Yang-Mills theory, but more broadly: it can serve as a guide to tell us which symbol formalisms are physically motivated.

Over the last few years, one of my driving concerns has been to understand what symbol alphabet formalism can look like beyond multiple polylogarithms. A symbol formalism has been proposed for elliptic multiple polylogarithms [14], inspired by a more general framework by mathematicians called the **motivic coaction** [19]. Working with a Master's student, I investigated whether the same approach could generalize beyond elliptic polylogarithms [20]. What we found was that there was not just one clear candidate, but many: a variety of different ways to construct symbol formalisms with very different results. Some of these formalisms gave symbols of wildly different lengths, longer or shorter "words" making manifest more or less of the functions' complexity. These formalisms may or may not make manifest the surprising properties of amplitudes in MSYM, but ideally they *should*: these properties likely have a physical origin, and thus they will be a powerful guide to finding a symbol that makes important physical properties manifest. Finally, it is not at all clear that all such symbols can be used consistently, with mathematicians expressing doubts that the method captures the structures made manifest by the motivic coaction.

My goal is to find a symbol alphabet for scattering amplitudes beyond polylogarithms, which promises to be a powerful tool for managing these challenging functions. As one can propose more than one such alphabet, the overall goal of this project is to find an alphabet that is *useful*. To find one, we will demand that such an alphabet:

1. Is mathematically consistent,
2. Captures as much as possible of the complexity of the functions in scattering amplitudes,
3. And makes manifest the physical basis of the surprising properties of symbols in planar $\mathcal{N} = 4$ super Yang-Mills.

A.1 State of the Art

A.1.1 Symbology Beyond Polylogarithms

For multiple polylogarithms, the symbol can be thought of as the simplest case of a more general mathematical object called a coaction. While the symbol breaks down a polylogarithm completely into individual letters encoding branch cuts, the coaction breaks the function into functions of intermediate complexity. The length of a function's symbol corresponds to its **transcendental weight**, a measure of the function's complexity. One can denote different components of the coaction by a pair, (n, m), giving the weights of the pieces that it decomposes a function into. The full coaction then encodes all possible decompositions of this type, giving extensive information about the analytic properties of the function. In this proposal, we will be interested in developing both symbol and coaction formalisms when appropriate, but will often restrict to mentioning just the symbol and its alphabet.

Currently, the only symbol formalism to see broad use for functions beyond multiple polylogarithms is the symbol for elliptic polylogarithms defined in ref. [14]. This reference also defined a coaction formalism for these functions. The elliptic symbol formalism was used to find ways to re-express contributions to string theory amplitudes in ref. [21] and three-loop contributions to a QCD parameter called ρ in ref. [22], as well as to analyze an elliptic one-loop amplitude in anti-de Sitter space in ref. [23]. Ref. [24] applied the formalism to the ten-particle elliptic double-box diagram in $\mathcal{N} = 4$ super Yang-Mills, the first elliptic diagram one encounters in this theory. They found interesting parallels to symbology for the theory's polylogarithmic amplitudes, with several branch cuts directly analogous to ones found in the polylog case. Ref. [25] then built off of these results to bootstrap the symbol and $(2, 2)$ coaction of the twelve-particle elliptic double-box (finding it without computing the diagram first, based on physically-motivated constraints), again in $\mathcal{N} = 4$ super Yang-Mills. Recently, ref. [26] has provided an algorithm for integrating elliptic polylogarithms at the symbol level.

Ref. [27] used the elliptic symbol formalism to investigate several identities between elliptic polylogarithms, finding an alternate derivation of a set of relations called the elliptic Bloch relations[28, 29] based on the symbol and certain relations between iterated integrals. Ref. [30] looked at more general relations of this kind, and defined a "symbol-prime" formalism that makes them manifest: in effect, introducing a new variant of the symbol formalism for relations between symbol letters themselves.

The symbol for elliptic polylogarithms was inspired by the general formalism of the motivic coaction [19], which provides a decomposition for generic period integrals (that is, integrals of algebraic functions over domains defined by polynomials, a very general class of interesting integrals), including integrals beyond elliptic polylogarithms. For several reasons, these formalisms are not identical, as can be straightforwardly seen by comparing the expressions for the coaction of the generic-mass two-loop sunrise integral, an important and common test case, in both formalisms [30, 31]. The coaction for elliptic polylogarithms is built around the iterated integral structure of these functions, in which one can express the two-loop sunrise diagram provided one divides out by a transcendental prefactor. It has a notion of transcendental weight consistent with degenerations to multiple polylogarithms around a particular kinematic point. In contrast, the motivic coaction acts on the sunrise integral without dividing out by a transcendental function. Whether the result can be interpreted in terms of iterated integrals is then less clear. The motivic coaction has a notion of weight that can jump discontinuously when degenerating to multiple polylogarithms, and which comes from a definition by Deligne in the context of the mathematical subject of mixed-Hodge theory [32].

The symbol and coaction for elliptic polylogarithms are built off of a particular iterated integral representation; equivalently one can think of them as built off of a division of functions into a choice of two parts, one, called unipotent, obeying a linear differential equation with a nilpotent connection matrix, and the other designated semi-simple. (The latter philosophy was employed to find a basis of consistent weight in ref. [33] for an integral related to a K3 variety, albeit one that can be expressed purely in terms of elliptic polylogarithms.) In my work with my student Andreas Forum [20] we applied this construction to a different iterated integral formulation that had previously been found for the higher-loop sunrise integrals [34], using it to derive a symbol and coaction formalism for these integrals. To date, this is the only explicit symbol or coaction proposed for a function beyond elliptic multiple polylogarithms.

A.1.2 Epsilon-Factorized Differential Equations Beyond Polylogarithms

When a scattering amplitude is evaluated in $d - 2\epsilon$ dimensions in the context of dimensional regularization, it can be expanded as a Laurent series in ϵ. It has been conjectured that the coefficient of ϵ^k has a maximum transcendental weight of $2L + k$ for L-loop Feynman integrals in four dimensions, and this has been proven for certain special cases in ref. [35]. This, concretely, is how we expect the form of Feynman integrals to represent their complexity: ideally, they should be expressible in iterated integrals with length that increases with loop order and with higher terms in the ϵ expansion.

As there are potentially many different ways to express a given Feynman diagram in terms of iterated integrals, there are many different ways to construct a symbol and coaction formalism. However, only some of these have the property that iterated integrals become longer at higher orders in the ϵ expansion. This property is guaranteed if the iterated integrals are solutions to a differential equation in ϵ-factorized form [17]. For Feynman diagrams evaluating to multiple polylogarithms, this form straightforwardly yields their expression in terms of multiple polylogarithms to any order in ϵ desired.

Beyond multiple polylogarithms, there have been a few proposals for forms of this kind. The first few such proposals focused on the elliptic case, where ref. [36] found an epsilon-factorized form for the equal-mass sunrise diagram, followed by ref. [37]'s result for the unequal-mass sunrise diagram. Ref. [38] proposed a general method for finding such a form in the elliptic case, based on finding a basis for which the period matrix is diagonal. Ref. [39] then compared the two approaches, finding that an intuitive goal for physicists, that an elliptic integral restricted to a polylogarithmic kinematic limit should have consistent weight, cannot be globally achieved.

One can loosely think of the difference between the approaches of the earlier refs. [36, 37] and the later ref. [38] as between "geometric" and "cut-based" approaches. Geometric approaches, like the first two references, leverage geometric descriptions of the varieties that appear in the problem, for example the modular properties of elliptic curves. Cut-based approaches use properties that one can more straightforwardly read off from the Feynman integrals, such as the result of localizing them on residues and closing their integration contours, for example the period matrix mentioned above. Cut-based approaches then can be more straightforwardly generalized, but tend to yield messier and less "canonical" results. One example of this is the observation in ref. [39] that the cut-based approach does not have consistent weight in any of its polylogarithmic limits.

Later approaches can then be classified in the same way. Ref. [40] found an ϵ-factorized form for the (elliptic) unequal-mass two-loop sunrise and made preliminary steps towards finding such a form for the (beyond elliptic, involving a K3 variety) unequal-mass three-loop sunrise. Their approach was geometric in character, invoking modular properties of the elliptic curve. Refs. [41, 42]'s approach was also geometric, explicitly inspired by refs. [36, 37], and focused on several Feynman integrals involving elliptic curves. Ref. [43]'s approach to the problem is potentially quite general, and represents a mix of the two types of approach. Beginning with a cut-based approach, they then make an ansatz for the entries of a differential equation inspired by the results of previous geometric approaches. They also provide computer code to implement their algorithm.

Beyond a single elliptic curve, ref. [44] found an ϵ-factorized form for an integral involving two elliptic curves with a geometric approach. Refs. [45, 46, 47] followed a geometric approach to go beyond the elliptic case. Beginning with the equal-mass three-loop sunrise, they continued to four loops, then generalized to any loop order based on a notion of Calabi-Yau operators [48, 49, 50, 51, 52, 53, 54, 55, 56]. Ref. [57], in contrast, demonstrated a "cut-based" approach that is in principle quite general, showing examples not only on elliptic integrals but also on diagrams involving more complicated algebraic varieties. Their approach was, however, not fully algorithmic, and required a healthy dose of human involvement in order to obtain a well-behaved result.

A.1.3 Multivariate Families of Calabi-Yau Varieties

A.1.4 Symbol Mysteries of $\mathcal{N}=4$ super Yang-Mills

There are three principal unsolved mysteries in the symbols of polylogarithmic amplitudes in planar $\mathcal{N}=4$ super Yang-Mills, each building off the other.

Scattering amplitudes in planar $\mathcal{N}=4$ super Yang-Mills have a symmetry referred to as dual conformal symmetry, in which the momenta of particles in the amplitude can be interpreted as the positions of cusps of a polygonal Wilson loop, a dual description in which the theory's conformal invariance is only broken by the presence of these cusps [58, 59, 60, 61, 62, 63]. This symmetry is anomalous, broken at loop level in a well-understood way in terms of the previously discovered Bern-Dixon-Smirnov ansatz [64], which captures all four- and five-particle amplitudes and is corrected for six or more particles by functions invariant under dual conformal symmetry [65, 66]. As the structure of the rest of the amplitude is known, in this proposal when I mention amplitudes in this theory I will generally mean only this remaining dual-conformal invariant piece.

(The dual-conformally invariant parts of) scattering amplitudes in planar $\mathcal{N}=4$ super Yang-Mills have been extensively investigated for six and seven particles (see the review in ref. [67] and references therein), with increasing attention paid to eight particles [68, 69, 70] and a few results at two loops for general numbers of particles [71, 72].

Theses all-n results led to the first unsolved mystery of symbols of amplitudes in this theory, the observation that the symbol letters of the amplitude correspond to coordinates (specifically, $\mathcal{A}$-variables) of a mathematical structure called a cluster algebra [73]. Later, based on observations of amplitudes with seven and eight particles, the second unsolved mystery was found. Cluster coordinates, when they appear in symbols of amplitudes, are restricted: two letters can only be adjacent in the symbol if they appear in the *same* cluster of the cluster algebra [74, 75, 76, 77, 78]. While there is a cluster algebra naturally associated to the dual conformally-invariant kinematic space appropriate to these amplitudes, neither of these properties follows inevitably from that fact: it is not obvious why symbol letters ought to be cluster coordinates, or why adjacent letters should be constrained to belong to the same cluster. And yet, these properties have been consistently observed everywhere where they straightforwardly apply.

The "straightforwardly" in the preceding sentence hides a caveat. Cluster coordinates are rational functions of an initial set of seed coordinates, but eight-particle amplitudes can contain letters depending on more general algebraic functions, beginning with square roots [72, 69], with higher roots expected to appear later [79, 80, 81]. At the same time, the cluster algebras appropriate to eight-particle amplitudes, unlike those for six and seven particles, have an infinite number of coordinates, suggesting that some generalization would be needed regardless.

There are various attempts in the literature to understand and predict these algebraic symbol letters, which can be loosely grouped into two categories. The first category of approaches attempts to find an appropriate generalization for the cluster algebra structure itself. One popular approach of this kind invokes tropical geometry, which offers a way to interpret an infinite sequence of cluster coordinates as limiting to an algebraic function [82, 83, 84, 85, 86]. A related approach using somewhat different mathematics was proposed in ref. [87]. More recently, both cluster algebra structures and a notion of cluster adjacency have been seen to emerge in examples from a more general notion, that of a fan structure associated to Gröbner bases [88].

The second category of attempts approaches the problem from the bottom up, computing the singularities of amplitudes in the theory and looking for patterns that could explain its cluster behavior. The natural starting point for such an approach is to attempt to solve the Landau equations, which describe all possible singularities of a scattering process [89]. In planar $\mathcal{N}=4$ super Yang-Mills the Landau equations have been employed directly to predict symbol alphabets in various contexts [90, 91, 92, 93, 94, 95, 96]. Interestingly, this method failed to predict certain symbol letters observed in the eight-particle amplitude at two loops [72], which is very surprising as the Landau equations are expected to characterize singularities of amplitudes completely.

It is less surprising when the Landau analysis suggests symbol letters that are not actually present. The Landau equations are applied diagram-by-diagram, and do not take into account the coefficients of those diagrams or possible cancellations between them. As such, recent investigations for seven particles found letters suggested by the Landau analysis that do not appear to be present in MSYM [96]. Full Landau analysis can also be quite cumbersome, and thus has so far been difficult to generalize to larger numbers of particles or loops. For both of these reasons, a number of more heuristic approaches have sprung up. These approaches tend to use concepts specific to MSYM, and thus try to make manifest special cancellations specific to that theory, but do not fully solve all of the Landau equations, typically just solving for solutions in which all particles are on-shell. Approaches of this kind have been pursued based on plabic graphs [97, 98, 99] and more recently on a method referred to as Schubert analysis [68, 100, 101, 102, 103, 104, 105]. The latter method is particularly notable as it has seen some success in predicting elliptic symbol letters as well [25].

Thus, the first two mysteries of symbols of polylogarithmic amplitudes in planar $\mathcal{N} = 4$ super Yang-Mills still stand decisively unresolved. The third is if anything more mysterious. In ref. [106], it was found that a certain three-point form-factor in this theory was dual to a kinematic limit of the six-particle amplitude under a map referred to as the **antipode**. This **antipodal duality** is especially mysterious because the antipode map does not seem to have any plausible physical meaning: it reverses the order of the symbol, exchanging branch cuts on the principal sheet for branch cuts on the furthest sheet. Later, it was found that this duality can be explained because both the (kinematically restricted) six-particle amplitude and the three-point form-factor are limits of a four-point form-factor, which itself possesses an antipodal self-duality (so is mapped to itself under the antipode map) [107]. This self-duality however remains unexplained.

A.2 Objectives

A.2.1 Map the Space of Consistent Symbols Beyond Polylogarithms

My work has revealed the existence of a wide variety of possible symbol alphabet formalisms beyond polylogarithms, each with different properties. My first objective will be to characterize the space of such formalisms, checking whether they are all mathematically consistent and finding a list of functions which allow one to specify a given symbol formalism.

A.2.2 Find Canonical Differential Equations Beyond Polylogarithms

The utility of symbol alphabets is closely linked to the idea of **canonical differential equations**, in which iterated integrals can be systematically built up out of an alphabet of letters appearing explicitly in the differential equation. My second objective will be to find differential equations of this type for Feynman diagrams beyond polylogarithms, and especially for diagrams involving Calabi-Yaus. A symbol formalism must be consistent with differential equations of this type if it is to exhibit the full complexity we expect out of Feynman integrals.

A.2.3 Understand $\mathcal{N} = 4$ Symbol Mysteries Beyond Polylogarithms

Most of the surprising properties of polylogarithmic symbols in MSYM remain unproven. In order to understand their origin, we will need to find physical principles behind them that have the potential to generalize beyond the space of polylogarithms. My third goal will be to find such principles, which will have the dual benefit of explaining ongoing mysteries in this theory and in providing a strong guide to which symbol alphabets beyond polylogarithms make manifest these types of important physical principles.

B Methodology

B.1 WP1: Mathematical Consistency

This work package will involve a translation effort between the language of mathematics and physics. In discussions between the team's physicists and mathematician, we will attempt to phrase the elliptic symbol formalism in the language of the motivic coaction. More specifically, we will investigate these questions:

1. Is the elliptic symbol formalism fully consistent? Are there situations in which applying it in different ways can give contradictory results? The fact that the elliptic symbol gives different answers when different functions are factored out of the formalism does not by itself indicate a contradiction, as this is an intended feature of the formalism. Rather, the contradictions we will look for will be ones in which the elliptic symbol asserts something different when used in different ways, even when it is supposed to give a unique answer. There are areas in which a mathematician may be worried that something of this sort happens,

related to its representation of monodromies and its assignment of transcendental weight. To investigate, we will look for situations in these areas where we can generate a counterexample, where using the elliptic symbol in two different ways gives two different answers. If no such situations exist, it should be possible to prove that the elliptic symbol formalism is consistent in these ways. Going further, if the formalism appears to be consistent we will show that the elliptic symbol of a function can be extracted in a consistent way from the motivic coaction.

2. Does the elliptic symbol formalism capture everything the motivic coaction does? If so, it should be possible to reconstruct the motivic coaction of an integral from its elliptic symbol (up to possibly fixing some constant terms). If it appears that the elliptic symbol indeed captures all relevant information about the function, we will show how to generate the motivic coaction of the function given its elliptic symbol. If instead the elliptic symbol does not capture everything the motivic coaction does, then we will demonstrate this, either by finding two functions that are identical under the elliptic symbol but have different motivic coactions or by finding an identity between functions that is manifest at the level of the motivic coaction but not in the elliptic symbol.

3. Are the concerns of the previous two steps relevant *in practice*? That is, if there is a contradiction in the elliptic symbol formalism, is it actually relevant to calculation of scattering amplitudes, or is it a problem that can be easily avoided in practical contexts? If the elliptic symbol formalism omits information present in the motivic coaction, do we expect the extra information to be useful, and do we expect it to be easily fixed by other considerations?[1] Are the motivic coaction and elliptic symbol both comparably computationally efficient and easy to use, or does one have an advantage in either area? While the first two questions are purely mathematical, this one will benefit greatly from dialogue between the mathematician on the team and the team's physicists.

Once we have understood the relationship between the elliptic symbol and motivic coaction in this way, we will have a strong foundation to propose new symbol alphabet formalisms for Feynman diagrams beyond elliptic polylogarithms. We will know what the requirements are for such formalisms to be consistent, and have a picture of what aspects of the integrals they do and do not capture. From this starting point, we will aim to characterize the space of consistent symbol alphabet formalisms. This will likely involve finding a list of functions which, when specified, determines a given symbol formalism uniquely.

We will then search this space of functions for symbol formalisms with interesting and useful properties, paying particular attention to those which are more useful for physics. This will involve feedback from the other work packages, as the requirements of canonical differential equations and the physical content of the symbol alphabet of MSYM will both offer clues as to which formalisms will be most useful. As the last element in this work package, we will develop computer code to implement the more useful notions of symbol alphabet for these functions, making this code freely available with a `Mathematica` interface so that it is broadly accessible to the community.

B.2 WP2: Capturing Complexity

B.3 WP3: Learning from MSYM

Our initial goal in this work package will be to develop a robust understanding of why symbol alphabets in planar $\mathcal{N} = 4$ super Yang-Mills have some of the surprising properties they do: principally, why symbol letters belong to cluster algebras and why they satisfy a cluster adjacency condition. A robust understanding in particular entails an understanding that is not restricted to multiple polylogarithms, but arises from general physical principles and thus can be applied to more general classes of functions.

Currently, the strongest lead on an understanding of this type are the results linking symbol alphabets to Gröbner fans of ref. [88], so these results will be our starting point. In particular, ref. [88] showed that the observed symbol letters and their adjacency relations both can naturally emerge from considering the structure of the space of polynomials in kinematic invariants (specifically, Plücker coordinates) modulo the relations those invariants satsify.

This derivation was impressive, but it does not by itself constitute a proof that symbol letters in this theory must have the relationship they do to cluster algebras. It relies on several assumptions about the nature of the letters. This can be realized intuitively by noticing that one could have in principle phrased the entire question in terms of kinematic invariants that make manifest all relations, for example an algebraically independent set of cross-ratios invariant under the conformal and dual-conformal symmetries of the amplitude. In those variables, one may need to consider algebraic symbol letters even for simpler cases (and thus the structure of Gröbner

[1] As an example of this in another context, some elliptic polylogarithm formalisms allow one to write functions that are not invariant under changing the sign of the square root defining the elliptic curve. As Feynman integrals are always invariant under this change of sign, this omission does not impact the usefulness of the formalism for our purposes.

bases may not be relevant in the same way). One also can add arbitrary constants to a letter and still obtain letters that are consistent under the symmetries of the problem.

Our approach will be to try to make these assumptions explicit. We will deform the "seed" expressions used to construct the Gröbner fan in various ways, computing the resulting Gröbner fans using appropriate mathematical software, likely `Macaulay2`. By testing these possibilities, we will narrow down a list of explicit requirements that together uniquely give the cluster structure observed in symbol alphabets in MSYM, and understand what those requirements mean physically.

Whatever new requirements suffice, they will need to have some explanation in terms of the expected singularities of the theory itself, and more specifically in terms of analysis of the Landau equations. As such, our next step will be to investigate whether the structure of the Landau equations imply the restrictions we found. Such an approach will ideally be able to directly prove that symbol alphabets in MSYM must correspond to cluster algebras and obey cluster adjacency, by showing that the physical requirements that lead to these properties are direct consequences of the singularity structure of the theory's Feynman diagrams.

Finally, because we will have identified physical conditions that do not depend on polylogarithms, we will then be in an ideal place to investigate their implications for amplitudes beyond polylogarithms. We will begin by investigating their consequences for the ten-particle elliptic double-box, and in particular observe what the equivalent property of cluster adjacency should be for its elliptic symbol. Bringing in the results of WP1, we will be able to propose a modified elliptic symbol alphabet formalism to make this property manifest if the original elliptic symbol alphabet formalism does not. Moving further in the project, we will be able to characterize what properties amplitudes involving Calabi-Yaus should have in order to be consistent with the physical principles we have uncovered, and constrain symbol alphabets for these functions to also have these properties.

B.4 The Team

In the course of this project, I will hire two PhD students and three Postdocs.

The **first PhD student** will begin by focusing on the third work package, the investigation of symbol letters in MSYM. Their project will begin by computing Gröbner fans using the software Macaulay2. From my experience mentoring advanced Master's students, students benefit from starting with a concrete programming task that they can use to develop mathematical intuition. A student beginning in this way should be well prepared to participate in the later stages of the project, interacting with the project's Postdocs and exploring new directions as they mature intellectually.

The **first Postdoc** will be a mathematician with expertise in the motivic coaction. They will be concerned with the first work package, and will work in close collaboration with me to characterize the relationship between the motivic coaction and the elliptic symbol formalism. The blend of rigor from their background and my focus on developing a useful formalism for physicists will yield strong results, ideas that are both well-justified and well-explained. In addition, they will be well-positioned to aid the first PhD student, as there will be overlap in their expertise and the algebraic geometry concepts that will be relevant to the third work package. This will both give the PhD student additional support and a lively community to discuss with, and give the postdoc valuable mentorship experience.

The **second Postdoc** will have expertise in Calabi-Yaus, possibly from researching them in the context of scattering amplitudes but ideally from experience with the string theory literature. They will focus on the second work package, where they will compute quantities from the Calabi-Yau literature that are essential to finding a canonical form for differential equations for diagrams involving Calabi-Yaus.

The **third Postdoc** will have experience with phenomenological calculations at scale, either in collider physics or in gravitational wave physics. They will overlap with the other Postdocs and PhD students, and their role will be to bring together the results from the three work packages into a usable and practical whole. To that end they will write computer code implementing our new symbol formalism and tools to find canonical differential equations, and it will be their responsibility to make sure this code is sufficiently optimized to be a practical tool for collider and gravitational wave physics.

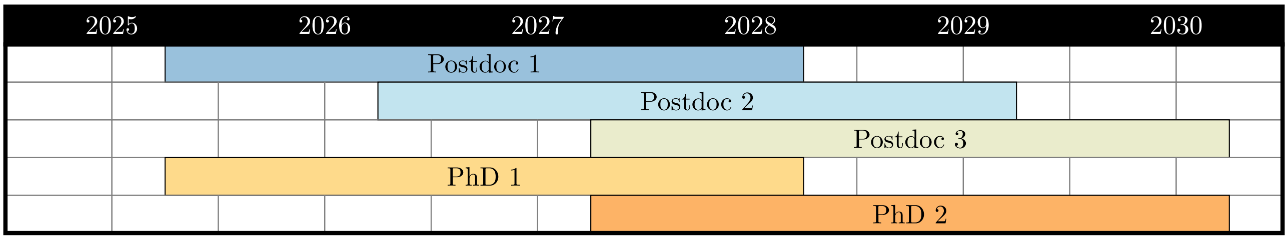
The **second PhD student** will also focus on preparing computer code at scale. They will initially do this in the context of MSYM, where they will extend the third work package to more challenging amplitudes with more loops and legs, potentially bootstrapping new quantities. As their work will concern both symbol formalisms and heavier-duty programming, they will also collaborate with the third postdoc on implementing our group's symbol formalism as useful computer code. Once again, this initial focus on computation reflects my experience mentoring advanced students, and the overlap with the third Postdoc offers opportunities both to impart important skills to the second PhD student and to give the third postdoc experience as a mentor that will be of value to their future career.

This project's host, the Institut de Physique Théorique of CEA Paris-Saclay, will be a lively research environment to host this project. The IPhT hosts many researchers interested in gravitational wave physics and in collider physics, as well as a sizable contingent of mathematical physicists who interact regularly with mathematicians. This research environment will provide stimulating discussions and useful feedback for both the work of this project and the postdocs and student hired. The IPhT has successfully hosted several ERC projects and is thus well prepared to host another.

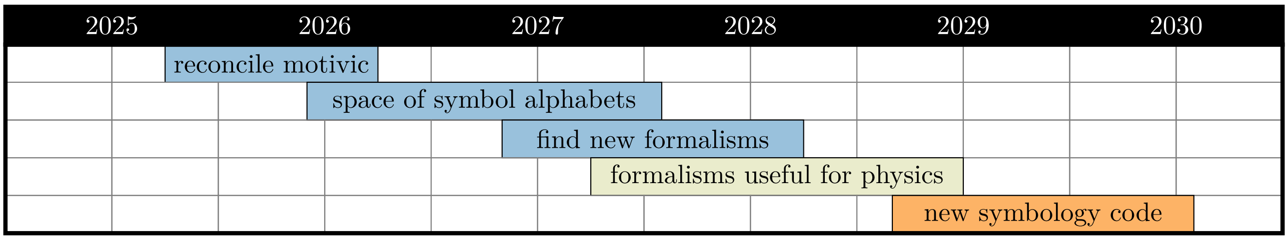
B.5 Timeline

The tables below locate the lines of research and personnel discussed above in the overall 60-month timeline of the project. Projects are color-coded with respect to which team member will be primarily responsible, with other colors denoting a mix of team members.

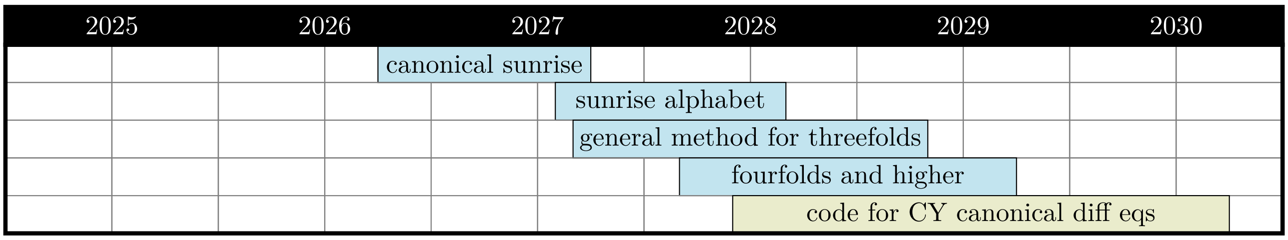
B.5.1 Team



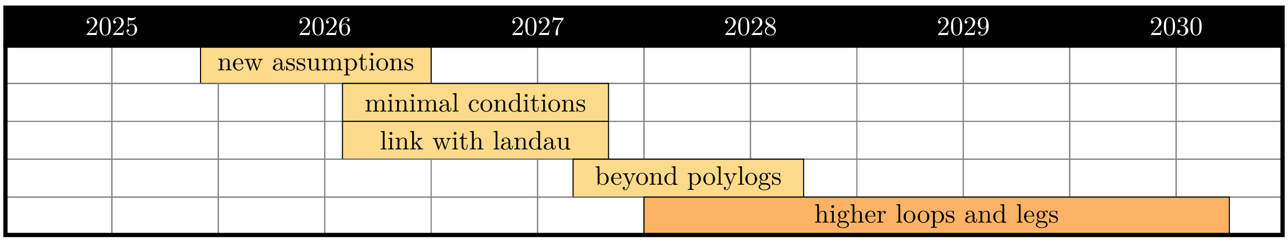
B.5.2 WP1: Mathematical Consistency



B.5.3 WP2: Capturing Complexity



B.5.4 WP3: Learning from MSYM



C Impact

By this project's end, we will have constructed a new formalism for computing scattering amplitudes in quantum field theories, one able to take the successes of the symbol for polylogarithms and bring them forward to the increasingly elliptic and Calabi-Yau state of the art. By pursuing mathematical consistency, faithful representation of complexity, and the clues in MSYM, we will establish a solid foundation for future investigations and a powerful research tool, useful from some of the smallest to the largest distance scales. Work on the motivic coaction earlier in the project will open a bridge to the mathematics community, while the development of computer code towards the end will provide practical tools for state of the art collider and gravitational wave physics.

A tool's greatness lies in what it enables. As we enter the era of precision physics, one of our strongest duties is to make sure we are making full use of all data we have access to. An ability to efficiently handle Feynman diagrams beyond polylogarithms is a necessary component to this goal. If we keep our eyes open, we can hope to learn something truly new: to find physics beyond the Standard Model, for example, or to revolutionize our models of neutron stars and black holes.

This project will establish me as a mentor and research leader, one capable not only of strong individual and collaborative work but with the vision to propose bigger goals and take the lead in achieving them. It will position my group as experts in the our formalism and of central value to the community, leading to new prospects for collaboration across Europe and the wider world.

References

[1] A. I. Davydychev and R. Delbourgo, "A Geometrical Angle on Feynman Integrals," *J. Math. Phys.* **39** (1998) 4299–4334, arXiv:hep-th/9709216 [hep-th].

[2] N. Arkani-Hamed and E. Y. Yuan, "One-Loop Integrals from Spherical Projections of Planes and Quadrics," arXiv:1712.09991 [hep-th].

[3] R. Boughezal *et al.*, "Theory Techniques for Precision Physics – Snowmass 2021 TF06 Topical Group Report," arXiv:2209.10639 [hep-ph].

[4] J. L. Bourjaily *et al.*, "Functions Beyond Multiple Polylogarithms for Precision Collider Physics," in *Snowmass 2021*. 3, 2022. arXiv:2203.07088 [hep-ph].

[5] J. L. Bourjaily, A. J. McLeod, M. von Hippel, and M. Wilhelm, "A (Bounded) Bestiary of Feynman Integral Calabi-Yau Geometries," *Phys. Rev. Lett.* **122** (2019) no. 3, 031601, arXiv:1810.07689 [hep-th].

[6] P. Lairez and P. Vanhove, "Algorithms for minimal Picard–Fuchs operators of Feynman integrals," *Lett. Math. Phys.* **113** (2023) no. 2, 37, arXiv:2209.10962 [hep-th].

[7] C. F. Doran, A. Harder, E. Pichon-Pharabod, and P. Vanhove, "Motivic geometry of two-loop Feynman integrals," arXiv:2302.14840 [math.AG].

[8] P. Lairez, E. Pichon-Pharabod, and P. Vanhove, "Effective homology and periods of complex projective hypersurfaces," arXiv:2306.05263 [math.AG].

[9] A. Buonanno, M. Khalil, D. O'Connell, R. Roiban, M. P. Solon, and M. Zeng, "Snowmass White Paper: Gravitational Waves and Scattering Amplitudes," in *Snowmass 2021*. 4, 2022. arXiv:2204.05194 [hep-th].

[10] M. Levi, "Effective field theories of post-newtonian gravity: A comprehensive review," arXiv:1807.01699 [hep-th].

[11] Z. Bern, J. Parra-Martinez, R. Roiban, M. S. Ruf, C.-H. Shen, M. P. Solon, and M. Zeng, "Scattering Amplitudes and Conservative Binary Dynamics at $\mathcal{O}(G^4)$," *Phys. Rev. Lett.* **126** (2021) no. 17, 171601, arXiv:2101.07254 [hep-th].

[12] J. Brödel, C. Duhr, F. Dulat, and L. Tancredi, "Elliptic Polylogarithms and Iterated Integrals on Elliptic Curves. Part I: General Formalism," *JHEP* **05** (2018) 093, arXiv:1712.07089 [hep-th].

[13] J. Brödel, C. Duhr, F. Dulat, and L. Tancredi, "Elliptic Polylogarithms and Iterated Integrals on Elliptic Curves II: an Application to the Sunrise Integral," *Phys. Rev.* **D97** (2018) no. 11, 116009, arXiv:1712.07095 [hep-ph].

[14] J. Brödel, C. Duhr, F. Dulat, B. Penante, and L. Tancredi, "Elliptic Symbol Calculus: from Elliptic Polylogarithms to Iterated Integrals of Eisenstein Series," *JHEP* **08** (2018) 014, arXiv:1803.10256 [hep-th].

[15] J. Brödel, C. Duhr, F. Dulat, B. Penante, and L. Tancredi, "Elliptic Feynman Integrals and Pure Functions," arXiv:1809.10698 [hep-th].

[16] A. B. Goncharov, M. Spradlin, C. Vergu, and A. Volovich, "Classical Polylogarithms for Amplitudes and Wilson Loops," *Phys. Rev. Lett.* **105** (2010) 151605, arXiv:1006.5703 [hep-th].

[17] J. M. Henn, "Multiloop Integrals in Dimensional Regularization Made Simple," *Phys. Rev. Lett.* **110** (2013) no. 25, 251601, arXiv:1304.1806 [hep-th].

[18] J. L. Bourjaily, Y.-H. He, A. J. Mcleod, M. von Hippel, and M. Wilhelm, "Traintracks Through Calabi-Yaus: Amplitudes Beyond Elliptic Polylogarithms," *Phys. Rev. Lett.* **121** (2018) no. 7, 071603, arXiv:1805.09326 [hep-th].

[19] F. Brown, "Notes on Motivic Periods," arXiv:1512.06410 [math.NT].

[20] A. Forum and M. von Hippel, "A Symbol and Coaction for Higher-Loop Sunrise Integrals," arXiv:2209.03922 [hep-th].

[21] C. R. Mafra and O. Schlotterer, "All Order α' Expansion of One-Loop Open-String Integrals," *Phys. Rev. Lett.* **124** (2020) no. 10, 101603, arXiv:1908.09848 [hep-th].

[22] S. Abreu, M. Becchetti, C. Duhr, and R. Marzucca, "Three-loop contributions to the ρ parameter and iterated integrals of modular forms," *JHEP* **02** (2020) 050, arXiv:1912.02747 [hep-th].

[23] S. F. Stawinski, "An Elliptic One-Loop Amplitude in Anti-de-Sitter Space," arXiv:2309.15059 [hep-th].

[24] A. Kristensson, M. Wilhelm, and C. Zhang, "Elliptic Double Box and Symbology Beyond Polylogarithms," *Phys. Rev. Lett.* **127** (2021) no. 25, 251603, arXiv:2106.14902 [hep-th].

[25] R. Morales, A. Spiering, M. Wilhelm, Q. Yang, and C. Zhang, "Bootstrapping Elliptic Feynman Integrals Using Schubert Analysis," *Phys. Rev. Lett.* **131** (2023) no. 4, 041601, arXiv:2212.09762 [hep-th].

[26] S. He and Y. Tang, "Algorithm for symbol integrations for loop integrals," *Phys. Rev. D* **108** (2023) no. 4, L041702, arXiv:2304.01776 [hep-th].

[27] J. Broedel and A. Kaderli, "Functional relations for elliptic polylogarithms," *J. Phys. A* **53** (2020) no. 24, 245201, arXiv:1906.11857 [hep-th].

[28] D. Zagier and H. Gangl, "Classical and elliptic polylogarithms and special values of l-series," in *The Arithmetic and Geometry of Algebraic Cycles*, Gordon, Brent B. and Lewis, James D. and Müller-Stach, Stefan and Saito, Shuji and Yui, Noriko, ed., pp. 561–615. Springer Netherlands, 2000.

[29] S. J. Bloch, *Higher regulators, algebraic K-theory, and zeta functions of elliptic curves*, vol. 11. CRM Monograph Series, 2000.

[30] M. Wilhelm and C. Zhang, "Symbology for elliptic multiple polylogarithms and the symbol prime," *JHEP* **01** (2023) 089, arXiv:2206.08378 [hep-th].

[31] M. Tapušković, "The cosmic Galois group, the sunrise Feynman integral, and the relative completion of $\Gamma_1(6)$," arXiv:2303.17534 [math.AG].

[32] C. A. M. Peters and J. H. M. Steenbrink, *Mixed Hodge Structures*, vol. 52. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics, 2010.

[33] J. Broedel, C. Duhr, F. Dulat, R. Marzucca, B. Penante, and L. Tancredi, "An analytic solution for the equal-mass banana graph," *JHEP* **09** (2019) 112, arXiv:1907.03787 [hep-th].

[34] K. Bönisch, C. Duhr, F. Fischbach, A. Klemm, and C. Nega, "Feynman integrals in dimensional regularization and extensions of Calabi-Yau motives," *JHEP* **09** (2022) 156, arXiv:2108.05310 [hep-th].

[35] H. S. Hannesdottir, A. J. McLeod, M. D. Schwartz, and C. Vergu, "Implications of the Landau equations for iterated integrals," *Phys. Rev. D* **105** (2022) no. 6, L061701, arXiv:2109.09744 [hep-th].

[36] L. Adams and S. Weinzierl, "The ε-form of the differential equations for Feynman integrals in the elliptic case," *Phys. Lett. B* **781** (2018) 270–278, arXiv:1802.05020 [hep-ph].

[37] C. Bogner, S. Müller-Stach, and S. Weinzierl, "The unequal mass sunrise integral expressed through iterated integrals on $\overline{\mathcal{M}}_{1,3}$," *Nucl. Phys. B* **954** (2020) 114991, arXiv:1907.01251 [hep-th].

[38] H. Frellesvig, "On epsilon factorized differential equations for elliptic Feynman integrals," *JHEP* **03** (2022) 079, arXiv:2110.07968 [hep-th].

[39] H. Frellesvig and S. Weinzierl, "On ε-factorised bases and pure Feynman integrals," arXiv:2301.02264 [hep-th].

[40] M. Giroux and A. Pokraka, "Loop-by-loop differential equations for dual (elliptic) Feynman integrals," *JHEP* **03** (2023) 155, arXiv:2210.09898 [hep-th].

[41] X. Jiang, X. Wang, L. L. Yang, and J. Zhao, "ε-factorized differential equations for two-loop non-planar triangle Feynman integrals with elliptic curves," *JHEP* **09** (2023) 187, arXiv:2305.13951 [hep-th].

[42] X. Jiang, X. Wang, L. L. Yang, and J. Zhao, "ε-forms for non-planar triangles with elliptic curves at two loops," arXiv:2309.07786 [hep-th].

[43] C. Dlapa, J. M. Henn, and F. J. Wagner, "An algorithmic approach to finding canonical differential equations for elliptic Feynman integrals," *JHEP* **08** (2023) 120, arXiv:2211.16357 [hep-ph].

[44] H. Müller and S. Weinzierl, "A Feynman integral depending on two elliptic curves," *JHEP* **07** (2022) 101, arXiv:2205.04818 [hep-th].

[45] S. Pögel, X. Wang, and S. Weinzierl, "The three-loop equal-mass banana integral in ε-factorised form with meromorphic modular forms," *JHEP* **09** (2022) 062, arXiv:2207.12893 [hep-th].

[46] S. Pögel, X. Wang, and S. Weinzierl, "Taming Calabi-Yau Feynman integrals: The four-loop equal-mass banana integral," arXiv:2211.04292 [hep-th].

[47] S. Pögel, X. Wang, and S. Weinzierl, "Bananas of equal mass: any loop, any order in the dimensional regularisation parameter," arXiv:2212.08908 [hep-th].

[48] M. Bogner, "Algebraic characterization of differential operators of Calabi-Yau type," arXiv:1304.5434 [math.AG].

[49] D. R. Morrison, "Picard-Fuchs equations and mirror maps for hypersurfaces," *AMS/IP Stud. Adv. Math.* **9** (1998) 185–199, arXiv:hep-th/9111025.

[50] A. Ceresole, R. D'Auria, S. Ferrara, W. Lerche, and J. Louis, "Picard-Fuchs equations and special geometry," *Int. J. Mod. Phys. A* **8** (1993) 79–114, arXiv:hep-th/9204035.

[51] G. Almkvist and W. Zudilin, "Differential equations, mirror maps and zeta values," arXiv:math/0402386.

[52] G. Almkvist, C. van Enckevort, D. van Straten, and W. Zudilin, "Tales of Calabi–Yau equations," arXiv:math/0507430.

[53] Y. Yang and W. Zudilin, "On sp4 modularity of picard-fuchs differential equations for calabi-yau threefolds," in *Gems in Experimental Mathematics*, T. Amdeberhan, L. A. Medina, and V. H. Moll, eds., vol. 517 of *Contemporary Mathematics*. 2010.

[54] M. Bogner and S. Reiter, "On symplectically rigid local systems of rank four and Calabi-Yau operators," arXiv:1105.1136.

[55] D. van Straten, "Calabi–Yau Operators," arXiv:1704.00164.

[56] P. Candelas, X. De La Ossa, and D. Van Straten, "Local Zeta Functions From Calabi-Yau Differential Equations," arXiv:2104.07816 [hep-th].

[57] L. Görges, C. Nega, L. Tancredi, and F. J. Wagner, "On a procedure to derive ϵ-factorised differential equations beyond polylogarithms," *JHEP* **07** (2023) 206, arXiv:2305.14090 [hep-th].

[58] J. Drummond, J. Henn, V. Smirnov, and E. Sokatchev, "Magic Identities for Conformal Four-Point Integrals," *JHEP* **0701** (2007) 064, arXiv:hep-th/0607160.

[59] Z. Bern, M. Czakon, L. J. Dixon, D. A. Kosower, and V. A. Smirnov, "The Four-Loop Planar Amplitude and Cusp Anomalous Dimension in Maximally Supersymmetric Yang-Mills Theory," *Phys. Rev.* **D75** (2007) 085010, arXiv:hep-th/0610248 [hep-th].

[60] Z. Bern, J. Carrasco, H. Johansson, and D. Kosower, "Maximally Supersymmetric Planar Yang-Mills Amplitudes at Five Loops," *Phys. Rev.* **D76** (2007) 125020, arXiv:0705.1864 [hep-th].

[61] L. F. Alday and J. M. Maldacena, "Gluon Scattering Amplitudes at Strong Coupling," *JHEP* **06** (2007) 064, arXiv:0705.0303 [hep-th].

[62] Z. Bern *et al.*, "The Two-Loop Six-Gluon MHV Amplitude in Maximally Supersymmetric Yang-Mills Theory," *Phys. Rev.* **D78** (2008) 045007, arXiv:0803.1465 [hep-th].

[63] J. Drummond, J. Henn, G. Korchemsky, and E. Sokatchev, "Dual Superconformal Symmetry of Scattering Amplitudes in $\mathcal{N}=4$ super Yang-Mills Theory," *Nucl. Phys.* **B828** (2010) 317–374, arXiv:0807.1095 [hep-th].

[64] Z. Bern, L. J. Dixon, and V. A. Smirnov, "Iteration of Planar Amplitudes in Maximally Supersymmetric Yang-Mills Theory at Three Loops and Beyond," *Phys. Rev.* **D72** (2005) 085001, arXiv:hep-th/0505205.

[65] L. F. Alday and J. Maldacena, "Comments on Gluon Scattering Amplitudes via AdS/CFT," *JHEP* **11** (2007) 068, arXiv:0710.1060 [hep-th].

[66] J. Bartels, L. N. Lipatov, and A. Sabio Vera, "BFKL Pomeron, Reggeized gluons and Bern-Dixon-Smirnov amplitudes," *Phys. Rev. D* **80** (2009) 045002, arXiv:0802.2065 [hep-th].

[67] S. Caron-Huot, L. J. Dixon, J. M. Drummond, F. Dulat, J. Foster, O. Gürdoğan, M. von Hippel, A. J. McLeod, and G. Papathanasiou, "The Steinmann Cluster Bootstrap for $N = 4$ Super Yang-Mills Amplitudes," *PoS* **CORFU2019** (2020) 003, arXiv:2005.06735 [hep-th].

[68] S. He, Z. Li, and C. Zhang, "Two-loop octagons, algebraic letters and $\bar{Q}$ equations," *Phys. Rev. D* **101** (2020) no. 6, 061701, arXiv:1911.01290 [hep-th].

[69] Z. Li and C. Zhang, "The three-loop MHV octagon from $\overline{Q}$ equations," *JHEP* **12** (2021) 113, arXiv:2110.00350 [hep-th].

[70] S. He, Z. Li, and C. Zhang, "A nice two-loop next-to-next-to-MHV amplitude in $\mathcal{N} = 4$ super-Yang-Mills," *JHEP* **12** (2022) 158, arXiv:2209.10856 [hep-th].

[71] S. Caron-Huot, "Superconformal symmetry and two-loop amplitudes in planar N=4 super Yang-Mills," *JHEP* **12** (2011) 066, arXiv:1105.5606 [hep-th].

[72] S. He, Z. Li, and C. Zhang, "The symbol and alphabet of two-loop NMHV amplitudes from $\bar{Q}$ equations," arXiv:2009.11471 [hep-th].

[73] J. Golden, A. B. Goncharov, M. Spradlin, C. Vergu, and A. Volovich, "Motivic Amplitudes and Cluster Coordinates," *JHEP* **1401** (2014) 091, arXiv:1305.1617 [hep-th].

[74] J. Drummond, J. Foster, and Ö. Gürdoğan, "Cluster Adjacency Properties of Scattering Amplitudes in $\mathcal{N} = 4$ Supersymmetric Yang-Mills Theory," *Phys. Rev. Lett.* **120** (2018) no. 16, 161601, arXiv:1710.10953 [hep-th].

[75] J. Drummond, J. Foster, and Ö. Gürdoğan, "Cluster Adjacency Beyond MHV," *JHEP* **03** (2019) 086, arXiv:1810.08149 [hep-th].

[76] J. Golden, A. J. McLeod, M. Spradlin, and A. Volovich, "The Sklyanin Bracket and Cluster Adjacency at All Multiplicity," *JHEP* **03** (2019) 195, arXiv:1902.11286 [hep-th].

[77] J. Mago, A. Schreiber, M. Spradlin, and A. Volovich, "Yangian invariants and cluster adjacency in $\mathcal{N} = 4$ Yang-Mills," *JHEP* **10** (2019) 099, arXiv:1906.10682 [hep-th].

[78] J. Mago, A. Schreiber, M. Spradlin, and A. Volovich, "A note on one-loop cluster adjacency in $\mathcal{N} = 4$ SYM," *JHEP* **01** (2021) 084, arXiv:2005.07177 [hep-th].

[79] N. Arkani-Hamed, J. L. Bourjaily, F. Cachazo, A. B. Goncharov, A. Postnikov, and J. Trnka, *Grassmannian Geometry of Scattering Amplitudes*. Cambridge University Press, 2016.

[80] J. L. Bourjaily, "Positroids, Plabic Graphs, and Scattering Amplitudes in MATHEMATICA," arXiv:1212.6974 [hep-th].

[81] J. L. Bourjaily, C. Vergu, and M. von Hippel, "Landau Singularities and Higher-Order Roots," arXiv:2208.12765 [hep-th].

[82] J. Drummond, J. Foster, O. Gürdogan, and C. Kalousios, "Algebraic singularities of scattering amplitudes from tropical geometry," arXiv:1912.08217 [hep-th].

[83] N. Henke and G. Papathanasiou, "How tropical are seven- and eight-particle amplitudes?," *JHEP* **08** (2020) 005, arXiv:1912.08254 [hep-th].

[84] N. Arkani-Hamed, T. Lam, and M. Spradlin, "Non-perturbative geometries for planar $\mathcal{N} = 4$ SYM amplitudes," arXiv:1912.08222 [hep-th].

[85] J. Drummond, J. Foster, O. Gürdoğan, and C. Kalousios, "Tropical fans, scattering equations and amplitudes," *JHEP* **11** (2021) 071, arXiv:2002.04624 [hep-th].

[86] N. Henke and G. Papathanasiou, "Singularities of eight- and nine-particle amplitudes from cluster algebras and tropical geometry," *JHEP* **10** (2021) 007, arXiv:2106.01392 [hep-th].

[87] L. Ren, M. Spradlin, and A. Volovich, "Symbol alphabets from tensor diagrams," *JHEP* **12** (2021) 079, arXiv:2106.01405 [hep-th].

[88] L. Bossinger, J. M. Drummond, and R. Glew, "Adjacency for scattering amplitudes from the Gröbner fan," arXiv:2212.08931 [hep-th].

[89] L. Landau, "On Analytic Properties of Vertex Parts in Quantum Field Theory," *Nuclear Physics* **13** (1959) no. 1, 181 – 192. `http://www.sciencedirect.com/science/article/pii/0029558259901543`.

[90] T. Dennen, M. Spradlin, and A. Volovich, "Landau Singularities and Symbology: One- and Two-Loop MHV Amplitudes in SYM Theory," *JHEP* **03** (2016) 069, arXiv:1512.07909 [hep-th].

[91] T. Dennen, I. Prlina, M. Spradlin, S. Stanojevic, and A. Volovich, "Landau Singularities from the Amplituhedron," *JHEP* **06** (2017) 152, arXiv:1612.02708 [hep-th].

[92] I. Prlina, M. Spradlin, J. Stankowicz, S. Stanojevic, and A. Volovich, "All-Helicity Symbol Alphabets from Unwound Amplituhedra," *JHEP* **05** (2018) 159, arXiv:1711.11507 [hep-th].

[93] I. Prlina, M. Spradlin, J. Stankowicz, and S. Stanojevic, "Boundaries of Amplituhedra and NMHV Symbol Alphabets at Two Loops," *JHEP* **04** (2018) 049, arXiv:1712.08049 [hep-th].

[94] I. Prlina, M. Spradlin, and S. Stanojevic, "All-loop singularities of scattering amplitudes in massless planar theories," *Phys. Rev. Lett.* **121** (2018) no. 8, 081601, arXiv:1805.11617 [hep-th].

[95] L. Lippstreu, M. Spradlin, and A. Volovich, "Landau Singularities of the 7-Point Ziggurat I," arXiv:2211.16425 [hep-th].

[96] L. Lippstreu, M. Spradlin, A. Yelleshpur Srikant, and A. Volovich, "Landau Singularities of the 7-Point Ziggurat II," arXiv:2305.17069 [hep-th].

[97] J. Mago, A. Schreiber, M. Spradlin, and A. Volovich, "Symbol alphabets from plabic graphs," *JHEP* **10** (2020) 128, arXiv:2007.00646 [hep-th].

[98] J. Mago, A. Schreiber, M. Spradlin, A. Y. Srikant, and A. Volovich, "Symbol alphabets from plabic graphs II: rational letters," *JHEP* **04** (2021) 056, arXiv:2012.15812 [hep-th].

[99] J. Mago, A. Schreiber, M. Spradlin, A. Yelleshpur Srikant, and A. Volovich, "Symbol alphabets from plabic graphs III: n = 9," *JHEP* **09** (2021) 002, arXiv:2106.01406 [hep-th].

[100] S. He, Z. Li, Q. Yang, and C. Zhang, "Feynman Integrals and Scattering Amplitudes from Wilson Loops," *Phys. Rev. Lett.* **126** (2021) 231601, arXiv:2012.15042 [hep-th].

[101] Q. Yang, "Schubert problems, positivity and symbol letters," *JHEP* **08** (2022) 168, arXiv:2203.16112 [hep-th].

[102] S. He, Z. Li, R. Ma, Z. Wu, Q. Yang, and Y. Zhang, "A study of Feynman integrals with uniform transcendental weights and their symbology," *JHEP* **10** (2022) 165, arXiv:2206.04609 [hep-th].

[103] S. He, J. Liu, Y. Tang, and Q. Yang, "The symbology of Feynman integrals from twistor geometries," arXiv:2207.13482 [hep-th].

[104] Q. Cao, S. He, and Y. Tang, "Cutting the traintracks: Cauchy, Schubert and Calabi-Yau," *JHEP* **04** (2023) 072, arXiv:2301.07834 [hep-th].

[105] S. He, X. Jiang, J. Liu, and Q. Yang, "On symbology and differential equations of Feynman integrals from Schubert analysis," arXiv:2309.16441 [hep-th].

[106] L. J. Dixon, O. Gurdogan, A. J. McLeod, and M. Wilhelm, "Folding Amplitudes into Form Factors: An Antipodal Duality," *Phys. Rev. Lett.* **128** (2022) no. 11, 111602, arXiv:2112.06243 [hep-th].

[107] L. J. Dixon, O. Gürdoğan, Y.-T. Liu, A. J. McLeod, and M. Wilhelm, "Antipodal Self-Duality for a Four-Particle Form Factor," *Phys. Rev. Lett.* **130** (2023) no. 11, 111601, arXiv:2212.02410 [hep-th].